Subject: **Metal Casting and Solidification (OE – 4)**								
Program: **B. Tech in Metallurgical Engineering**				Subject Code: **MME0405**			Semester: **IV**	
Teaching Scheme (Hours per week)				**Examination Evaluation Scheme (Marks)**				
Lecture	**Tutorial**	**Practical**	**Credits**	**University Theory Examination**	**University Practical Examination**	**Continuous Internal Evaluation (CIE)-Theory**	**Continuous Internal Evaluation (CIE)-Practical**	**Total**
3	0	0	3	16/40	0	24/60	0	200

Course Objectives

1. To acquire the knowledge about the fundamentals of the casting, basic terminology related to casting process.
2. To make students aware about the alternative method for the manufacturing of component for engineering applications.

CONTENTS

UNIT-I

[10 hours]

Moulding materials: Properties, preparation & testing:
Functional requirements of moulding materials, moulding practice & special requirements of core sands, foundry sands & binders, sand preparation & systems, silica programme.

UNIT-II

[10 hours]

Production techniques:
Manufacture of sand castings, mould production, moulding techniques, cores & core making, melting & casting, finishing operation. Shell, investment and die casting process, centrifugal casting, other special techniques.

UNIT-III

[10 hours]

Liquid metals and gating & feeding of castings:
Fluidity of liquid metals, mould factors in metal flow, the gating of casting. The feeding characteristics of alloys, methods for feeding of castings; design modifications, padding chills & insulation, the feeding of cast irons, solidification modeling.

UNIT-III

[10 hours]

Casting design and analysis of casting defects:
Preliminary stages & production stages, process & alloy selection, physical design features, quality assessment & control. Categories of defects, gas defects, shrinkage defects, contraction defects, compositional errors & segregation, inclusions & sand defects, shaping faults.

Course Outcomes
1. To apply knowledge about how to manufacture the intricate casting what should be the process parameter, design of pattern, mould, etc.
2. To apply the theory about the melting practice of different cast alloy.
3. To apply the knowledge to overcome defects generated during casting.

Text Books
1. R. W. Heine, C. R. Loper and P. C. Rosenthal, "Principles of Metal Casting", Tata McGraw Hill, 2nd Edition, 2017, ISBN: 9780070993488.
2. P. L. Jain, "Principles of Foundry Technology", Tata McGraw Hill, 2nd Edition, 1987, ISBN: 9780074516980.

Reference Books
1. P. C. Mukherjee, "Fundamentals of Metal Casting Technology", Oxford & IBH, 1st Edition, 1988, ISBN: 9788120403635.
2. P. R. Beeley, "Foundry Technology", Butterworth-Heinemann, 2nd Edition, 2001, ISBN: 9780750645676.
3. H. F. Taylor and M. C. Flemings, "Foundry Engineering", Wiley Eastern, 1st Edition, 1959, ISBN: 9780471848431.
4. D. Kumar and S. K. Jain, "Foundry Technology", CBS Publications, 1st Edition, 2007, ISBN: 9788123902906.

Web Resources
1. NPTEL MOOC Course on "Principles of Casting Technology" (https://onlinecourses.nptel.ac.in/noc17_me11/preview)